

SAMUEL OYENEYE

Lagos, Nigeria

+234 916 831 9229 | samueloyeneye1@gmail.com | [Github](#) | [LinkedIn](#) | [Google Scholar](#) |

RESEARCH INTEREST

As a Machine Learning Researcher, I find it impactful to reduce the high costs of sophisticated GPUs, which are unevenly distributed globally, while ensuring privacy. My interest lies in developing scalable, resource-efficient architectures that minimise the complexity of AI systems, with a focus on trustworthiness. This includes exploring human neural capabilities to design sensitive, aware, and efficient algorithmic methods.

EDUCATION

BSc., Federal University of Agriculture, Abeokuta, Nigeria 2019 - 2024
Department of Computer Science

RESEARCH EXPERIENCE

Independent ML Researcher 2024 - present
ML Collective

- Co-Lead Investigator, **Privacy Isn't Free: Benchmarking the System Cost of Privacy-Preserving ML** 2025
 - **Advisor:** Steven Kolawole (CMU)
 - **Contribution:** Developed the YAML framework called PrivacyBench, which benchmarks the privacy-utility-cost of PPML techniques on CNN and Transformer baselines on medical image datasets. This research presents significant results on energy consumption, showing that a single or a combination of privacy techniques can cause, with clear metric highlights.
- Co-Lead Investigator, **Secure and Scalable Horizontal Federated Learning for Bank Fraud Detection** 2024
 - **Contribution:** Designed the HFL transformer architectures where we applied the technique on the BAF-base dataset, outperforming the majority techniques on the BAF comparison benchmark
- Co-Lead Investigator, **Understanding LLM Reasoning Capabilities through Rebus Puzzles** In progress
 - **Contribution:** Annotated and experimented on visual language models (VLMs) on a variety of tasks, ranging from zero-shot to few-shot, to investigate the performance of both proprietary and open source models on idiomatic rebus puzzles. The experiments highlight a clear gap in reasoning capabilities and Question-Answering of these models.
- Contributor, **Multimodal Agreement-Based Cascading (ABC)** In progress
 - **Advisor:** Steven Kolawole (CMU)
 - **Contribution:** Experimented using SGLang inference library with different multimodal LLMs of various sizes from 1B parameters to 40B+ parameters, on the MMLU benchmark, to apply the Agreement-Based Cascading efficiency method to see if we can prove that smaller models can achieve similarly significant results compared to bigger models

Machine Learning Researcher 2025
Cohere Labs, **Aya Expedition Project**

- Contributor, **Sparse Upcycling Aya Vision**
 - **Contribution:** Created an eval pipeline for the multimodal LlavaBench on the Aya Vision model. The research aims to reduce training FLOPs and improve convergence speed compared to conventional dense and traditional upcycling approaches on multimodal vision models

PUBLICATIONS S=In Submission, W=Workshop, * Indicates Equal Contribution

[W.1] Mahi Aliyu Aminu^{1*}, ..., ..., **Samuel Oyeneeye***. **Effect of Domain Generalization Techniques in Low-Resource Systems**. Under Review at EACL Student Research Workshop [2026]

- [S.1] Nnaemeka Obiefuna, **Samuel Oyeneye**, Similoluwa Odunaiya, Iremide Oyelaja, Steven Kolawole. **PRIVACYBENCH: Privacy Isn't Free in Hybrid Privacy-Preserving Vision Systems**. (Accepted at ICML ES-FOMO III Workshop [2025]). | [\[pdf \]](#)
- [W.1] Nnaemeka Obeifuna*, ..., ..., **Samuel Oyeneye***. **Secure and Scalable Horizontal Federated Learning for Bank Fraud Detection**. (ICLR Advances in Financial AI Workshop 2025) | [\[pdf \]](#)

SELECTED WORK EXPERIENCE

AI Engineer

2025 - present

Splexx Labs | Remote

- Architected and built the entire backend infrastructure of the Tuneiverse AI music platform, implemented 5 AI features for audio and video transformations using Meta Demucs model, OpenAI-whisper model, FFMPEG and other libraries
- Designed the deployment service of the machine learning model pipelines on Azure, with a latency reduction of 20% on audio and 14% on video data, with regular metric logging and evaluation monitoring on the Redis caching

Senior Data Scientist

2025

BCT Limited

Lagos, Nigeria

- Designed and deployed an AI-powered agentic balanced scorecard system for ICS Outsourcing, leveraging intelligent automation to track performance metrics across one SBU and provide real-time strategic insights
- Developed a comprehensive Educational Management Information System (EMIS) for school student management, automating administrative workflows and data-driven reporting
- Mentored 9 students in data science methodologies and supervised 1 intern over 3 months

Machine Learning Engineer (Contract), Remote

2024 - 2025

SDTCorp

California, US

- Fine-tuned Llama-3.1 model on robot-framework data using Unsloth, saving memory up to 70%, with no loss, achieving a 91% accuracy, then distilled to smaller sizes
- Deployed the model to AWS and integrated it into the software testing chatbot built with streamlit, with knowledge graph functionalities, incorporating RAG and Neo4j database

Python Engineer Intern

2023 - 2024

Vale Finance Limited

Lagos, Nigeria

- Developed a Telcom service provider classification model with an accuracy of 96%, integrated with FastAPI
- Optimised backend process with Docker, with real-time responses and classification, incorporated with batch processing and authentication of the model's API

LEADERSHIP AND COMMUNITY SERVICE

Reviewer, Deep Learning Indaba (Rwanda '25)

2025

- Selected and reviewed 27 applicants' essays and submissions

Machine Learning Tutor, Data Science Network, AI Invasion

2023

- Facilitated hands-on classes for over 50 participants across age ranges of teens, young adults and adults, for a 5-day beginner to advanced machine learning tutorial

AI/ML Mentor, Data Science Network, FUNAAB

2022 - 2023

- Prepared study curriculum and tutored 5 mentees immersively on machine learning internals to build real-world projects

Core Team Community Lead, Google Developer Student Club, FUNAAB

2021 -2022

- Organised online and offline events, including technical tutorials, workshops and bootcamps for over 3000 students, empowering them with skills and expertise in the tech industry

SKILLS

- **Programming Languages:** Python (proficient), Golang
- **Frameworks & Libraries:** PyTorch, Hugging Face Transformers, TensorFlow, Langchain, ADK (Agent Development Kit), Streamlit, Unsloth, Numpy, Pandas, Scikit-learn, Matplotlib, YAML, SGLang
- **Specialization:** Large Language Models (LLMs & VLMs), Finetuning, Knowledge Distillation, Optimization, Retrieval-Augmented Generation (RAG), Computer Vision (CNNs, ViT)
- **MLOps:** GCP, AWS, Render, Docker, FastAPI, YAML
- **Databases:** SQL (Postgres, MySQL and SQLite), GraphDB (Neo4j), VectorDB (Pinecone), MetricDB (Weight & Bias)
- **Tools:** Git & Github, Linux

AWARDS AND GRANTS

Simons Computational Neuroscience Imbizo Summer School (~5% success rate)	2025
Best Poster Award at Deep Learning Indaba	2025